6.H3 Justifying Similarity

Lesson Introduction

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<tr><td>Benchmark</td><td>MA.912.GR.2.9 Justify the criteria for triangle similarity using the definition of similarity in terms of non-rigid transformations.</td><td>Learning Target</td><td colspan="2">I can use the definition of similarity to justify that SSS similarity, SAS similarity, and AA similarity are sufficient to show that two triangles are similar.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">What makes a figure similar to another?
How can you justify that two figures are similar through transformations?</td></tr>
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Similarity in Terms of Non-Rigid Transformations
Similarity
<u>Additional Terms:</u>
Dilation
Translation
Reflection
Rotation
Congruent
Corresponding</td><td colspan="2">Patty Paper</td><td>Consider creating a rubric to evaluate students' responses to 6.H3.3 Wrap-Up.</td></tr>
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6.H3.2 Collaboration question 1a directly asks students to record what they notice. Allow students the opportunity to express some of their wonderings before moving on. Questioning students about what they wonder can pique their curiosity and increase engagement in the lesson.</td><td>MA.K12.MTR.2.1: Multiple Representations
The transformations in this lesson include visuals, written descriptions, and algebraic descriptions of the same transformations. Providing these multiple representations encourages students to demonstrate understanding and make connections.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The structure of this lesson highlights the critical features of transformations that produce similar figures. As students progress from identifying transformations to interpreting the algebraic representations of the transformations, they continually refer back to the fundamental similarity definitions and theorems. In the middle of this lesson, students are given the opportunity to incorporate patty paper to connect these transformations into a more kinesthetic learning process.</td></tr>
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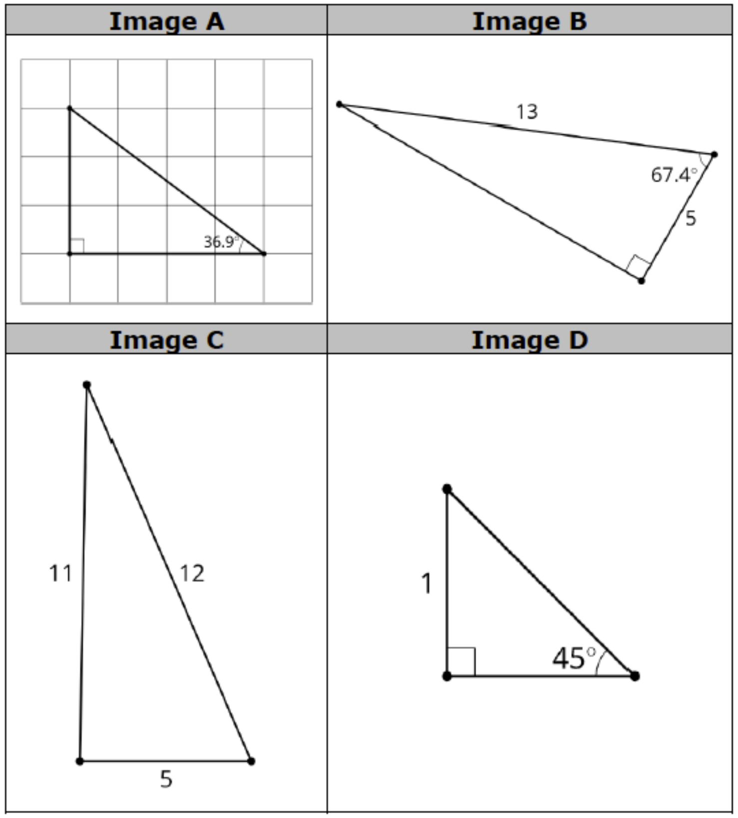
6.H3.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students select the image that they believe does not belong with the other three images and justify why.*

6.H3.1 Warm-Up: Which One Doesn't Belong?



1. Choose an image that does not belong and explain your choice.
   Answers will vary. A is different because it has a grid. B is different because it shows the lengths of two sides. C is different because it is not a right triangle. D is different because it is a $45^\circ, 45^\circ, 90^\circ$.

Students are asked to find which image doesn't belong. However, each shape may not belong for different reasons. This question is more about student's reasoning as to why the images do not belong.

Students may enjoy the opportunity to "convince" other students that they are correct.

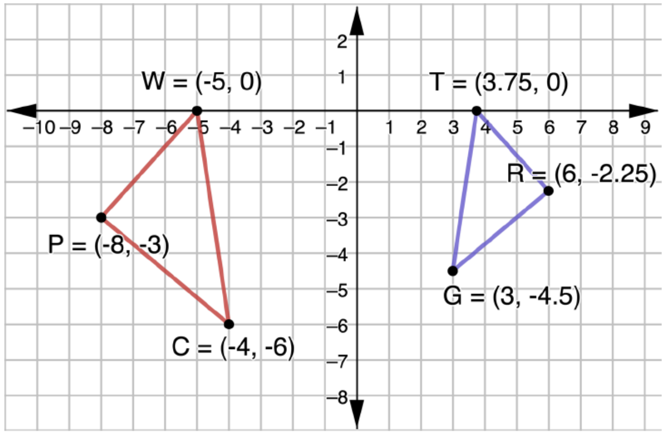
- *Have a few students explain what they selected and why.*
- *Pair up students who had different answers.*
- *Each student should make two to three solid points as to why they are correct (not why the other student is wrong). About* $20 - 30$ *seconds.*
- *Students can have "open discussion" for* 30 *seconds.*
- *Each student can decide to stay with their original answer or change. If they change, they should explain why.*

6.H3.2 Collaboration: Justifying Similarity

***Component Overview**: Students analyze a pair of triangles and determine what transformations map one to the other and what the relationship is between the corresponding parts. Through mapping and copying figures, students are led to justifying the definition of similarity in terms of non-rigid motions.*

6.H3.2 Collaboration: Justifying Similarity

1. ΔPWC and ΔRTG are shown.



a. What do you notice about the two triangles?
Answers will very. I notice ΔTRG is smaller and pointing in the other direction.

b. What transformations can be applied so that ΔWPC is mapped to ΔTRG?
Answers will vary. A reflection across the y-axis and then a dilation centered at the origin with a scale factor of $\frac{3}{4}$.

c. What can you conclude from the transformations?
ΔTRG is similar, but not congruent to ΔWPC.

Omario's and Ernest's answers to the question "If ΔPWC and ΔRTG are similar by angle-angle similarity, what information would need to be given?" are shown.

Omario	Ernest
$\angle WPC \cong \angle TRG$ and $\angle WCP \cong \angle TGR$	$\angle PWC \cong \angle RTG$ and $\angle PCW \cong \angle RGT$

Notice and Wonder
Before releasing students to complete this component through collaboration, consider presenting question 1a whole-group and extending it to include "What do you wonder?" Steer the conversation to statements wondering about whether the triangles are similar and how this could be verified.

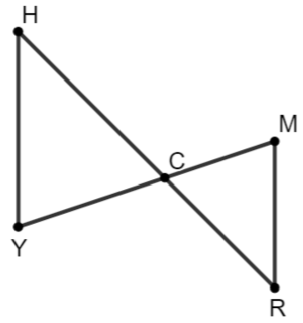
This component is purposefully scaffolded to build students' understanding and confidence as they progress through.

6.H3.2 Collaboration: Justifying Similarity (continued)

d. What do you notice about their answers?
I notice that both students are correct.

e. Explain why both students correctly answered the question.
Both students identified two pairs of congruent corresponding angles. Any two pairs of congruent corresponding angles will be enough to prove the triangles are similar.

2. ΔMRC and ΔYHC are shown.



a. Complete the table.

Given Information	True or False? $\Delta MRC \sim \Delta YHC$ by AA Similarity
$\angle HCY \cong \angle RCM$ $\angle H \cong \angle R$	○ True ○ False
$\angle HCY \cong \angle RCM$ $\frac{MR}{YH} = \frac{MC}{YC}$	○ True ○ False
$\angle H \cong \angle R$ $\angle Y \cong \angle M$	○ True ○ False

Students may think that SSA is justification for similarity. Remind students that when two sides are proportional and a pair of angles are congruent, the angle must be the included angle between the proportional sides. Consider showing a counterexample of two triangles that have SSA, but are not similar.

6.H3.2 Collaboration: Justifying Similarity (continued)

b. Select all of the transformations in a sequence that could be applied so that ΔMRC coincides with ΔYHC:
- ☐ rotate ΔMRC about point C
- ☐ rotate ΔMRC about point
- ☐ reflect ΔMRC across $\overline{MR}$
- ☐ reflect ΔMRC across $\overline{YH}$
- ☐ translate ΔMRC left and up
- ☐ dilate ΔMRC using point C as the center
- ☐ dilate ΔMRC using point R as the center

c. Discuss with your partner what is true about the triangles when all of the transformations have been applied.
Answers will vary. The triangles are congruent after all transformations are applied.

Similarity in terms of non-rigid transformations is one or more rigid transformations (reflection, rotation, translation) combined with a dilation that leads to two figures being similar.

Similarity is having exactly the same shape but not necessarily the same size. Two figures are similar if and only if one figure can be obtained from the other by a dilation, or a sequence of transformations that includes a dilation.

d. Use the definition of similarity in terms of non-rigid motions to justify that ΔMRC and ΔYHC are similar.
By rotating ΔMRC about point C and then dilating it using point C as the center of dilation, ΔMRC is mapped onto ΔYHC. Since the sequence of transformations includes a dilation, the definition of similarity in terms of non-rigid motions applies.

In addition to the answer shown, another sequence of transformations that could be applied is a rotation about point R, a translation left and up, and then a dilation using point R as the center.

While monitoring discussions, listen for the use of the vocabulary 'similar'.

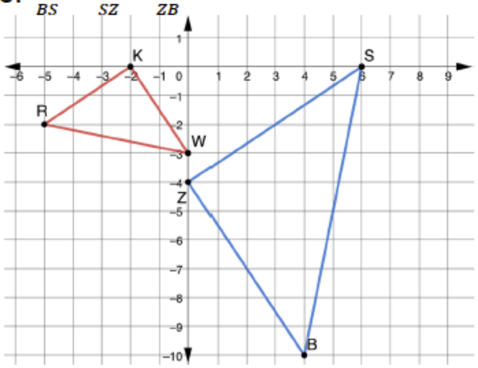
6.H3.2 Collaboration: Justifying Similarity (continued)

3. Consider the following statement.

If it is possible to find one or more transformations that will transform a figure such that all corresponding points coincide with the other figure, then the figures will be similar.

Explain what this statement means in your own words.
Answers will vary. If rigid or non-rigid motions are used to map one figure onto another then the shapes are similar.

4. ΔRKW and ΔBZS are shown.
Assume: $\frac{RW}{BS} = \frac{WK}{SZ} = \frac{KR}{ZB}$



a. Explain how you can use patty paper to verify $\Delta RKW \sim \Delta BZS$.
I can trace one of the triangles and then place each angle one at a time on top of the corresponding angle in the other triangle to determine that the angle measures are the same. Therefore, the triangles will be similar by AA similarity.

b. Write a sequence of transformations that will map ΔRKW onto ΔBZS so that $\Delta RKW \sim \Delta BZS$.
Answers will vary. Rotate ΔRKW 90° counterclockwise about the origin and then dilate the result using a scale factor of 2 with the origin as the center of dilation.

Three Reads
Consider implementing this routine by having the students read the statement three times and then discussing what the statement means prior to them writing an explanation of the meaning.

Read #1: Shared Reading centered around "what is this situation about?" (1 minute)

Read #2: Individual, Pairs, or Shared Reading centered around "what can be counted?" (3–5 minutes)

Read #3: Individual, Pairs, or Shared Reading centered around "how could we solve the problem?" (1–2 minutes)

As students are given the opportunity to develop the concept of transformations with a kinesthetic approach, the patty paper will confirm many students' suspicions about similar figures.

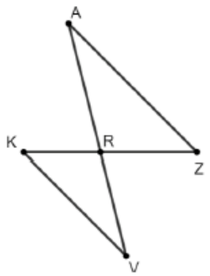
c. If it is given that ΔBZS is a result of $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (2x, 2y)$ to ΔRKW, are the triangles congruent?
No, they are not congruent.

d. If it is given that ΔBZS is a result of $(x, y) \rightarrow (-y, x)$ then $(x, y) \rightarrow (2x, 2y)$ to ΔRKW, are the triangles similar?
Yes, they are similar.

e. Discuss with your partner what criteria would have to be given so that SAS similarity would be shown as sufficient to justify triangles are similar. Summarize your discussion.
To justify triangles are similar using SAS similarity, two pairs of corresponding sides with an included angle would need to be given. The ratios of the pairs of corresponding sides would need to be equivalent and the included angles would need to be congruent.

While monitoring students, prompt them to give an explanation for why the triangles are congruent or similar in questions 4c and 4d along with their answer.

5. Katelyn, Diego, and Amelia are writing conclusions about proving similarity in triangles. They used ΔKRV and ΔZRA to write their conclusions. Their conclusions are shown.



Assume: $\angle VKR \cong \angle AZR$ and $\angle VRK \cong \angle ARZ$

6.H3.2 Collaboration: Justifying Similarity (continued)

Katelyn's Conclusion	Diego's Conclusion	Amelia's Conclusion
To prove similarity of two triangles, apply the definition of congruence in terms of rigid motions to show that two angles that are assumed to be congruent are congruent. This is sufficient to prove similarity of two triangles.	Show that two triangles are congruent using rigid motions and dilation. Then, apply the definition of similarity in terms of non-rigid transformations. This is sufficient to prove the triangles are similar.	By applying a series of rigid transformations and a dilation, it could be shown that the triangles are similar by showing that one set of sides are proportional and assumed two angles are congruent.

a. Whose conclusion do you agree with?
Diego is correct.

b. Explain why you agree with that student.
Answers will vary. Diego correctly uses the definition of similarity in terms of non-rigid transformations.

c. Pick one student whose conclusion is incorrect. Write a note to them to help them understand their error.

Answers will vary.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

Allow the students to elaborate about Amelia's mistake. Provide sentence stems, such as:

________ made a mistake when he/she _________ because_________.

_________ should have ___________ instead of ______________.

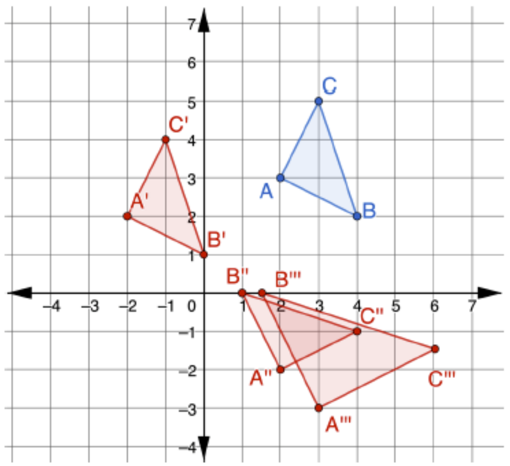
The next time _______ sees ______ he/she should ___________________.

6.H3.3 Wrap-Up: Cool Down

Component Overview: *Students perform a series of transformations and determine if the resulting figure is similar or congruent.*

6.H3.3 Wrap-Up: Cool Down

1. ΔABC is shown.



A sequence of three transformations is shown.

- $(x, y) \rightarrow (x - 4, y - 1)$
- $(x, y) \rightarrow (y, x)$
- $(x, y) \rightarrow (1.5x, 1.5y)$

a. Draw the result of each transformation.

b. Complete the table by noting if the preimage and image are congruent or similar.

Transformation	Relationship of Preimage to Image
$(x, y) \rightarrow (x - 4, y - 1)$	*Congruent*
$(x, y) \rightarrow (y, x)$	*Congruent*
$(x, y) \rightarrow (1.5x, 1.5y)$	*Similar*

Consider making a list of misconceptions to look for when reviewing student responses. Use these to help formulate a rubric for student responses. For example, students may be able to answer 1b correctly but make a mistake on question 1a. A student who accidently miscounts spaces on the first transformation but has all other aspects correct is not showing a lack of knowledge of the benchmark whereas a student work that shows the second transformation as a reflection across the x-axis is showing a misunderstanding.

Consider establishing your rubric and then scoring a few student answers and revising the rubric as necessary. Once your rubric is established choose a few different student responses and delete any identifying factors. Give students the rubric and 3-5 student responses and ask the students to use the rubric to provide a score for each response. Students enjoy evaluating work and using a rubric may help them consider what errors they made. If there is time, then have students share their scores for the common papers to see if students “graded” them the same.

This can also be used as a review activity at the end of the unit.

Use the data collected in 6.H3.3 Wrap-Up to determine any remediation steps needed.